ENRIQUE PAILLAS

Astrophysicist

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MOTIVATION

I am a researcher interested in the study of the large-scale structure of the Universe, with a special focus on its characterization and the extraction of physical information out of its properties. The distribution of galaxies is the result of a long-standing contest between the expansion of the Universe and the growth of cosmic structure. We believe the present-day *cosmic web* assembled in a hierarchical structure formation scenario, where small-scale quantum fluctuations from the early Universe grew over time due to gravitational instabilities. My main research line is devoted to using statistical tools to extract the cosmological information that is encoded in this cosmic web, by means of the analysis of galaxy clustering and weak gravitational lensing.

EXPERIENCE

Doctoral researcher

Pontificia Universidad Católica de Chile

Mar 2017 – Ongoing Santiago, Chile

- Development of an automated, fully parallel pipeline to identify and analyse cosmic voids in cosmological simulations and galaxy survey data.
- Identification and analysis of cosmic voids in the BOSS and EBOSS surveys, with the aim of constraining the growth of structure around regions with low galaxy density.

Research assistant

CIGIDEN

Sep 2019 – Feb 2020 Santiago, Chile

- Processing and analysis of Sentinel-2 satellite geo-spatial data for wildfire identification in Chile.
- Development of an automated pipeline for wildfire identification using machine learning and artificial intelligence on geo-spatial data.

Data science advisor

QUAOS

Jan 2018 – Jan 2019 Santiago, Chile

- Development of a pipeline to predict and manage bank remittances, based on the bank's historical transaction's data.
- Assessment study about the impact of Chilean's sulfuric acid production on the international market.

Research assistant

Universidad Andrés Bello

Jan 2010 – Feb 2015 Santiago, Chile

- Research under the supervision of Prof. Giuliano Pignata, calibrating astronomical photometric data of type II supernovae from the Chilean Automatic Supernovae Search project.
- Research under the supervision of Prof. Timo Anguita, calibrating the light-curves of gravitationally lensed quasars, using data taken with the GROND instrument.

TEACHING

Teaching assistant - Introduction to Astronomy

P. Universidad Católica de Chile

2018

Teaching assistant - Differential Calculus

Universidad Andrés Bello

2014

Teaching assistant - Differential Equations

Universidad Andrés Bello

2013

Teaching assistant - Introduction to Mathematics

Universidad Andrés Bello

2012

LANGUAGES

Spanish (native) ●●●●●
English ●●●●●

IT SKILLS

Python ●●●●●
Fortran 90 ●●●●●
C ●●○○○
SQL ●●○○○
Git ●●●○○
High-performance computing ●●●○○

RELEVANT COURSES

- Computational Astrophysics
- Cosmology and Dark Matter
- Probability and Statistics
- Astrostatistics
- Quantum Mechanics II
- Introduction to Electrodynamics
- Visualizing Geospatial data in Python
- Introduction to Git for Data Science
- Introduction to SQL for Data Science

PUBLICATIONS

Optimal void finders in weak lensing maps
Davies, Paillas et al., 2020, MNRAS, staa3262

The Santiago-Harvard-Edinburgh-Durham void comparison II: unveiling the Vainshtein screening using weak lensing
Paillas et al., 2018, MNRAS, 484, 1149

The evolution of the baryon fraction in haloes as a cause of scatter in the galaxy stellar mass in the EAGLE simulation
Kulier, A. et al. (including Paillas, E.), 2019, MNRAS, 482, 3261K

The Santiago-Harvard-Edinburgh-Durham void comparison I: SHEDding light on chameleon gravity tests
Cautun, Paillas et al., 2017, MNRAS, 476, 3195

DESCQA: An Automated Validation Framework for Synthetic Sky Catalogs
Mao, Y.-Y. et al. (including Paillas, E.), 2018, ApJ, 234, 36

Baryon effects on void statistics in the EAGLE simulation
Paillas et al., 2017, MNRAS, 470, 4434

SN 2009ib: A Type II-P Supernova with an Unusually Long Plateau
Takáts, K. (et al., including Paillas, E.), 2015, MNRAS, 450, 3137

SN2011hs: a Fast and Faint Type IIb Supernova from a Supergiant Progenitor
Bufano, F. (et al., including Paillas, E.), 2014, MNRAS, 439, 1807

SN 2009N: Linking normal and subluminous type II-P Sne
Takáts, K. et al. (including Paillas, E.), 2014, MNRAS, 438, 368

ACADEMIC VISITS

Academic visit to research about the impact of baryonic feedback on the properties of cosmic voids, under the supervision of Dr. John Helly
Institute for Computational Cosmology - Durham University - Durham, UK
Jan 2016 - Feb 2016

Academic visit to research about the modelling of redshift-space distortions around cosmic voids, under the supervision of Dr. Ariel Sánchez
Max Planck Institute for extraterrestrial Physics - Munich, Germany
May 2019 - Aug 2019

ORAL PRESENTATIONS

Constraining the growth rate using redshift-space distortions around voids
Mock Innsbruck: the connection between galaxies and dark matter haloes
Innsbruck, Austria, 2020

Weak lensing by voids as a test for alternative cosmological models
University Observatory, Ludwig-Maximilians-Universität München
Munich, Germany, 2019

Constraining modified gravity using weak lensing by voids
Dynamics of the Large-scale Structure
Munich, Germany, 2019

Unveiling modified gravity using cosmic voids
Max Planck Institute for Astronomy
Munich, Germany, 2019

Unveiling cosmology with void regions
The Cosmic Web: From Galaxies to Cosmology
Edinburgh, UK, 2019

The Santiago-Harvard-Edinburgh-Durhan void comparison project
Mock Cordoba: galaxy formation for gravity and cosmology
Cordoba, Argentina, 2019

Unveiling modified gravity using cosmic voids
AAA-SOCHIAS binational meeting
La Serena, Chile, 2018

Using voids to constrain modified gravity
LSST DESC Collaboration meeting
San Francisco, Florida, USA, 2018

Baryon effects on void statistics
Latin American Regional IAU meeting
Cartagena de Indias, Colombia, 2016

Baryon effects on void statistics
Mock Santiago: Preparing for next generation surveys
Santiago, Chile, 2016